MILITARY SPECIFICATION

TEST SET, RADIO SET

AN/ARM-165

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for shop use and shall provide the capability to perform functional tests and fault isolation on AN/ARC-159 and AN/ARC-159(V) UHF Radio Set equipments. The Test Set shall provide the capability to fault isolate the major module subassemblies of the Radio Set under test.

1.2 Associated Equipment - The equipment shall operate with the following associated equipment: (See Para. 6.8).

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MS3057	Clamp, Cable, Electrical Connector
MS3102	Connector, Receptacle, Electric, Box Mounting
MS3106	Connector, Plug, Electric, Straight
MS3107	Connector, Plug, Electric, Quick Disconnect

FSC-6625

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SPECIFICATIONS, continued

Military

MIL-L-3661	Lampholder, Indicator Lights, Indicator-light Housings, and Indicator-light Lenses, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-S-8805	Switches and Switch Assemblies, Sensitive and Push, Snap Action, General Specification for
MIL-M-10304	Meter, Electrical Indicating, Panel Type, Ruggedized, General Specification for
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures, Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for Use with Electronic and Electrical Equipment, General Specification for
MS27473	Connector, Plug, Electric, Crimp Type, Straight, Bayonet, Coupling Series 1-a
MS27484	Connector, Plug, Electric, Crimp Type, Straight, Bayonet Coupling, Grounding Series 1-a
MS27506	Adapter, Electric Connector Series 1 and 1
MS3505E	Switch, Toggle One Pole, Sealed Toggle
MIL-C-39019	Circuit Breakers, Magnetic, Low Power, Sealed, Trip-Free, General Specification for

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SPECIFICATIONS, continued

Military

MIL-C-39024	Connector, Electrical, Test Point Type, Panel Type, Single Test Point. (Threaded). Low Voltage
MS75028	Switch, Toggle, One Pole, Unsealed
MS75029	Switch, Toggle, Two Pole, Unsealed

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Expoential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packing and Packaging of

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
AR-10	Maintainability of Avionics Equipment and Systems, General Requirement for
AR-21	Aeronautical Requirement Ground Support Equipment
AR-30	Integrated Maintenance Management for Aeronautical Weapons, Weapon Systems and Related Equipment

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2.1.1 Availability of Documents

(1) When requesting specifications, standards, drawings and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Testing - This specification makes provision for testing (see 4)

3.2 Materials, parts, and processes - The materials, parts, and processes shall be in accordance with MIL-T-21200.

3.2.1 Selection of parts, materials, and processes - Except as specified herein, MIL-T-21200 shall apply for the selection of parts, materials and processes.

3.2.2 Approval of nonstandard parts and materials - Approval for the use of nonstandard parts and materials shall be in accordance with MIL-T-21200.

3.3 Design and construction - The design and construction of the equipment shall be in accordance with MIL-T-21200

3.3.1 Total weight - The total weight of the equipment, including cables, and module extenders, shall be a minimum consistent with good design and shall not exceed 40 pounds

3.3.2 Microelectronic items - When used, microelectronic items shall be in accordance with AR-5.

3.3.3 Electron devices - Electron tubes and semiconductor devices shall be in accordance with MIL-T-21200.

3.3.4 Modules, maintenance - The electronic portions of the equipment shall be divided into maintenance modules as specified in AR-10 Modules shall normally be considered repairable.

3.3.5 Time totalizing meter - Not applicable.

3.3.6 Cables and connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-T-21200 Insofar as is practicable, all connectors on the front panel shall be mounted along the lower edge of the panel and in no case shall their location be such that the connecting cables will interfere with the operating controls

3.3.6.1 Cabling - The test equipment cabling shall be compatible with the equipment to be tested. It shall be the electrical equivalent of the cabling used in the normal installation of the equipment being tested.

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3.3.6.2 External connectors - External connectors shall provide for the following:

Designation	Type	Function
J1	MS3102R-28-21S	Mates Cable W1
J2	MS3102R-22-14P	Mates Cable W2
J3	MS3102R-24-28S	Mates Cable W3
J4	MS3102R-20-27S	Mates Cable W4
J5	MIL-C-39024	Mates low side of external DC voltmeter to primary Tested Unit power line
J6	MIL-C-39024	Mates high side of external DC voltmeter to primary Tested Unit power line
J7	MIL-C-39024	Mates low side of external DC voltmeter to 27.5V Sw output
J8	MIL-C-39024	Mates high side of external DC voltmeter to 27.5V Sw output
J9	MIL-C-39024	Mates low side of external DC voltmeter to REXMT CONT OUT line
J10	MIL-C-39024	Mates high side of external DC voltmeter to REXMT CONT OUT line
J11	MIL-C-39024	Mates low side of oscilloscope to DATA B
J12	MIL-C-39024	Mates high side of oscilloscope to DATA A
J13	MIL-C-39024	Mates low side of oscilloscope to CLOCK B
J14	MIL-C-39024	Mates high side of oscilloscope to CLOCK A
J15	MIL-C-39024	Mates oscilloscope trigger input to Data Word Sync line
J16	Dual Binding Post	Mates audio voltmeter to Headset output

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3.3.6.2 External connectors, continued

Designation	Type	Function
J17	Dual Binding Post	Mates audio voltmeter to Selected Audio Output
J18	Dual Binding Post	Mates audio voltmeter to output of the MIC Adapter
J19	Dual Binding Post	Mates audio signal generator to input of the MIC Adapter
J20	Dual Binding Post	Mates audio voltmeter to output of test set audio amplifier
J21	Dual Binding Post	Mates audio signal generator to input of test set audio amplifier
J22	Dual Binding Post	Mates Vacuum Tube Voltmeter to selected IF AGC line
J23	Banana Tip Jack, Black	Mates ammeter calibrator to (-) side of M1
J24	Banana Tip Jack, Black	Mates ammeter calibrator to (+) side of M1

3.3.7 Controls - Front panels controls layout and labeling shall be as in Figure 1 The detailed requirements of the controls shall be as specified in 3.7.5.

3.3.8 Human engineering - Human engineering shall be in accordance with MIL-T-21200.

3.4 Availability - The availability shall be a minimum of 99 percent when calculated using the results of the Reliability and Maintainability demonstrations of paragraphs 3 4.1 and 3 4.2 as follows

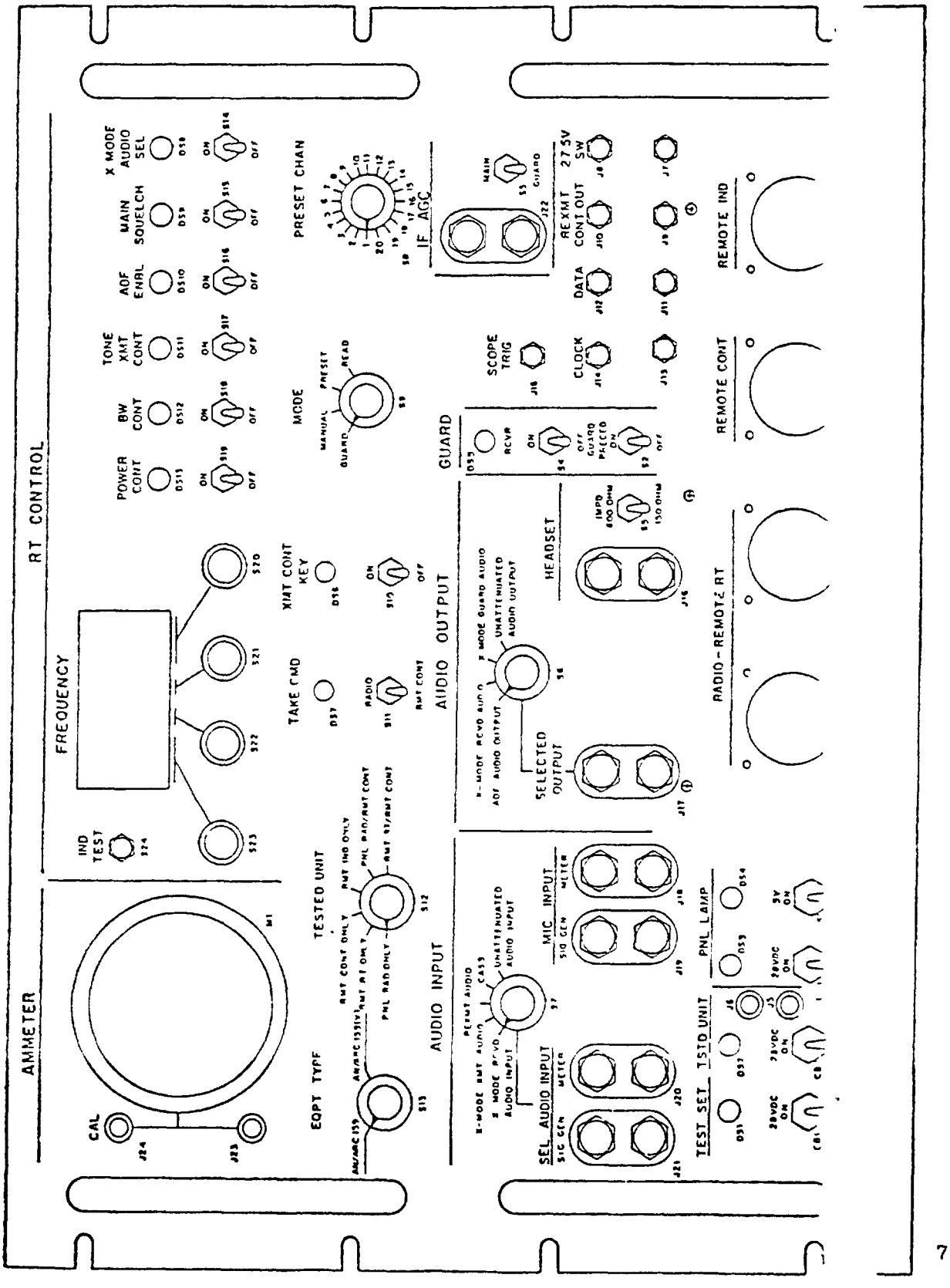
$$\text{Availability} = \frac{\theta}{\theta + \bar{M}_{ct}}$$

where:
θ is the observed Mean-Time-Between-Failure (MTBF) equal to the total operating time of the equipment divided by the number of failures, as defined in MIL-STD-781

and:
M_{ct} is equal to the Mean Corrective maintenance downtime, as defined in MIL-STD-471.

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FIGURE 1 - FRONT PANEL (TYPICAL)



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3.4.1 Reliability - The contractor shall conduct a reliability program in accordance with MIL-T-21200. The program plan shall be approved by the procuring activity.

3.4.1.1 Operating stability - The equipment shall operate as specified for not less than 2000 hours or 12 months, whichever comes first, without scheduled maintenance, realignment, or adjustment of any control not readily acessible to the operator during normal operation.

3.4.1.2 Operating life - The equipment shall have a minimum total operating life of 10,000 hours. Parts requiring scheduled replacement due to wear during the life of the equipment and, the wearout life of such parts, shall be determined by the contractor and submitted to the procuring activity

3.4.1.3 Reliability in Mean-Time-Between-Failure (MTBF) - The equipment, including any built-in-test provisions, shall have a specified 2000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4 4.3.

3.4.2 Maintainability - The contractor shall conduct a maintainability program in accordance with MIL-T-21200. The program plan shall be approved by the procuring activity Built-in-test features, modular construction and packaging, and provisions for test points shall be as specified in AR-10. The program will provide appropriate input data for Integrated Logistics support requirements in accordance with AR-21 or AR-30 applicable to the avionic equipment to be tested by this support equipment

3.4.2.1 Maintenance times - The equipment, including any built-in provisions, shall be designed to meet the following quantitive maintainability requirements for corrective maintenance when tested in accordance with 4.2.2.1.

(a) Mean time to repair, $\bar{M}_{ct}$ 2 hours

(b) Maximum time to repair, $M_{max\ ct}$ 4 hours

3.4.2.2 Maintenance manhours per maintenance action - The total maintenance manhours per maintenance action shall not exceed 5 hours when conducted in accordance with 4.2 2.1.

3.4.3 Probability of successful performance of specified test (Compatibility)- The probability of successful performance during a complete test (all test sequences) shall be a minimum of 96 percent when tested in accordance with the provisions of 4.2.2.2. The probability of successful performance of a complete test of the avionic equipment to be tested by this support equipment shall include (a) not detecting a failure in the avionic equipment to be testec, (b) indicating a non-existent failure in the avionic equipment to be tested, (c) the probability of causing a failure in the avionic equipment tc be tested, and (d) the probability that a failure in the avionic equipment under test will cause a failure in the support equipment (except for the activation of a protective device)

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3.4.4 Interference control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of MIL-STD-461.

3.4.5 Maintenance provisions and field testing - Provisions for maintenance shall be as specified in MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practical for ease of field testing and maintenance.

3.4.6 Nomenclature and nameplates - Nomenclature assignment and nameplates approval for equipment identification shall be in accordance with MIL-N-18307.

3.4.7 Service conditions - Service conditions (environmental) shall be in accordance with MIL-T-21200.

3.4.7.1 Altitude - The altitude environment shall be in accordance with MIL-T-21200, paragraph 3.2.19.3 for Class 3 equipment.

3.4.7.2 Shock - The shock environment shall be in accordance with MIL-T-21200, paragraph 3.2.19.5.

3.4.7.3 Vibration - The vibration environment shall be in accordance with MIL-T-21200, paragraph 3.2.19.4 for Class 3 equipment.

3.4.7.4 Humidity - The humidity environment shall be in accordance with MIL-T-21200, paragraph 3.2.19.2.

3.4.7.5 Salt atmosphere - Not applicable.

3.4.7.6 Sand and dust - Not applicable.

3.4.7.7 Temperature-altitude - The temperature-altitude environment shall be in accordance with MIL-T-21200, paragraph 3.2.19.1 for Class 3 equipment.

3.4.8 Standard conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient
Altitude	Normal Ground
Vibration	None
Humidity	Room Ambient
Input Power Voltage	28VDC ± 2VDC

3.5 Transportability - Transportability requirements shall be in accordance with MIL-T-21200.

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3.6 Performance characteristics - The performance of the equipment shall be as specified herein.

3.7 Detail requirements -

3.7.1 Function - The Test Set shall have the capability to interface and test the following equipments:

AN/ARC-159 Radio Set
AN/ARC-159(V) Radio Set
RT/1150/ARC-159(V) Receiver-Transmitter, Radio
C-9577/ARC-159(V) Control, Radio Set
ID-1972/ARC-159(V) Indicator, Frequency Channel

The Test Set shall be capable of interfacing and testing the AN/ARC-159(V) equipment individually or a Radio Set with a Control, with or without an Indicator, and a Receiver-Transmitter with a Control, with or without an Indicator.

The Test Set shall consist of a rack mountable panel with enclosed chassis, four (4) cables to interconnect the equipment(s) under test and six (6) module extenders. The Test Set shall be designed to be used with Common Ground Support Equipment listed in paragraph 6.9. Common Ground Support RF Equipment shall be connected directly to the UUT rather than connected through the Test Set.

To allow testing of the AN/ARC-159(V) frequency control serial data system, the Test Set shall contain frequency control selector switches, a preselect channel select switch, a digital frequency readout, a 1KHz clock and 32 bit frame 1KHz data generation and data receive circuitry. The Test Set shall not be required to contain circuitry to test the AN/ARC-159 Radio Set parallel data frequency control.

The Test Set shall provide the capability to operate the unit under test control panel lamp circuits.

3.7.2 Form factor - The Test Set Panel outline dimensions shall not exceed those as specified in Figure 2.

3.7.3 Weight - The weight of the Test Set, including extenders and cables, shall not exceed 40 pounds.

3.7.4 Content - The Test Set shall contain the following assemblies, subassemblies, circuits, cables and module extenders:

(1) Test Set Panel (Including integral chassis and covers)
  (a) Digital Display Assembly
  (b) Power Supply Assembly
  (c) Clock Generator
  (d) Data Generator
  (e) Data Receiver

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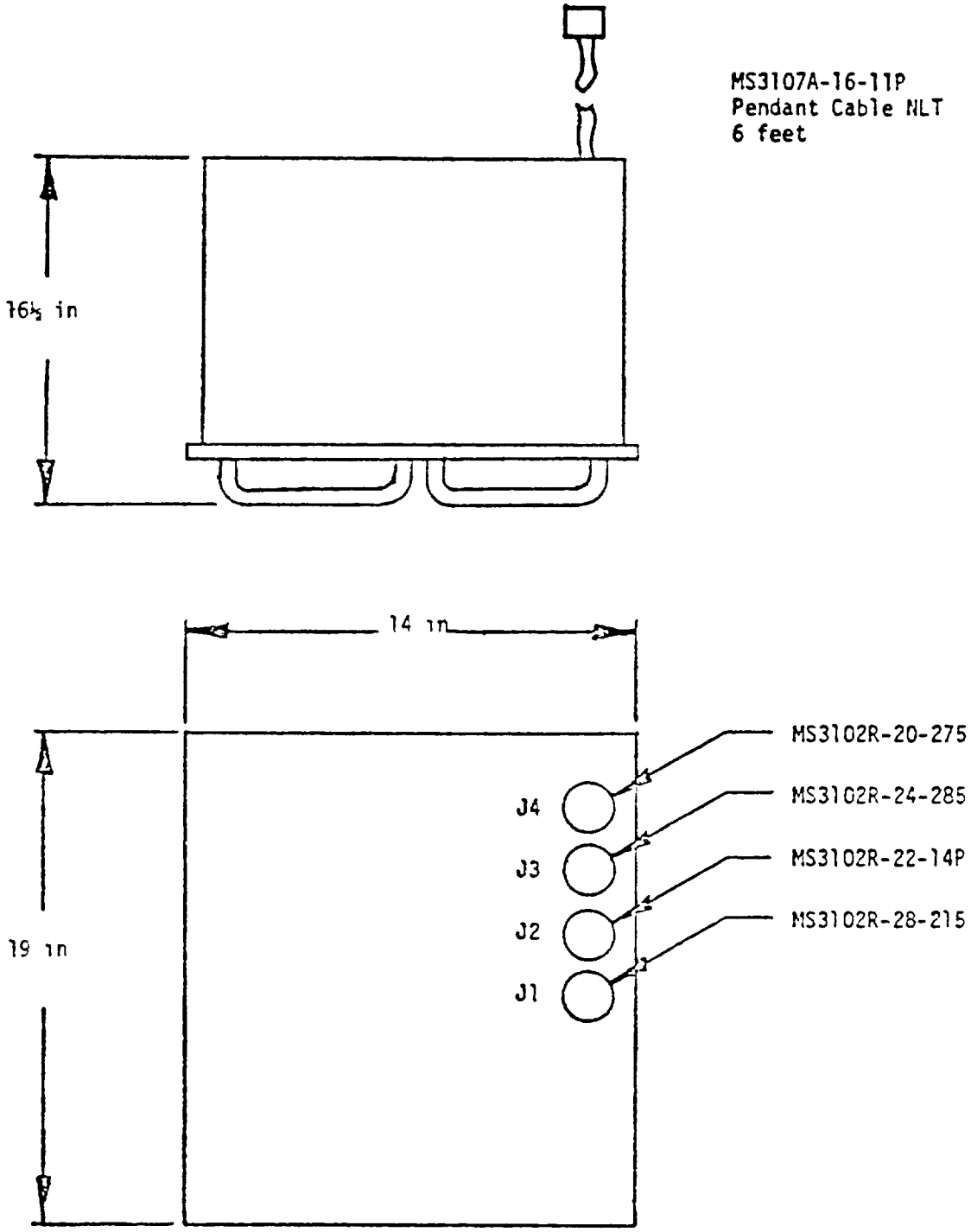


FIGURE 2 - OUTLINE DIMENSIONS

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3.7.4 Content, continued

(2) Power Amplifier/Power Supply Module Extender
(3) Control Module Extender
(4) Frequency Synthesizer Module Extender
(5) Translator Module Extender
(6) IF/AF Module Extender
(7) Guard Receiver Module Extender
(8) Cable Assembly, W1
(9) Cable Assembly, W2
(10) Cable Assembly, W3
(11) Cable Assembly, W4

3.7.5 Controls -

3.7.5.1 Test Set panel controls -

3.7.5.1.1 TEST SET circuit breaker, CB1 - Circuit breaker, CB1, shall control the primary DC power to the Test Set. The circuit breaker shall provide the capability of interrupting both sides of the primary DC power lines. CB1 shall conform to MIL-C-39019.

3.7.5.1.2 TESTED UNIT circuit breaker, CB2 - Circuit breaker, CB2, shall control the primary DC power to the unit(s) under test. The circuit breaker shall conform to MIL-C-39019.

3.7.5.1.3 28VDC PNL LAMP circuit breaker, CB3 - CB3 shall control the DC power to the unit(s) under test 28V panel lamps circuit. CB3 shall conform to MIL-C-39019.

3.7.5.1.4 5VDC PNL LAMP switch, S1 - Switch, S1, shall be a toggle switch to control the DC power to the unit(s) under test 5V panel lamps circuit. S1 shall conform to MS35058

3.7.5.1.5 GUARD PRECED switch, S2 - Switch, S2, shall be a toggle switch that shall be capable of placing a ground on the GUARD PRECEDENCE line of the Panel Radio or Remote R/T. S2 shall not be operational in the AN/ARC-159 position of the EQPT TYPE switch, S13, or in the RMT CONT ONLY and RMT IND ONLY positions of the TESTED UNIT switch, S12. S2 shall conform to MS75028.

3.7.5.1.6 HEADSET IMPD switch, S3 - Switch, S3, shall be a two position toggle switch which shall provide the capability of switching either the 600 Ohm or 150 Ohm headset output lines to the HEADSET test jacks, J16 S3 shall conform to MS75028.

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3.7.5.1.7 GUARD RCVR switch, S4 - Switch, S4, shall be a toggle switch that shall be capable of placing a ground on the GUARD RECEIVER ON-OFF line of the unit under test. S4 shall be operational only in the RMT R/T ONLY and RMT RT/RMT CONT positions of the TESTED UNIT switch, S12 S4 shall conform to MS75028.

3.7.5.1.8 IF AGC selector switch, S5 - Switch, S5, shall be a two position toggle switch which shall provide the capability of switching either the MAIN or GUARD IF AGC lines to the IF AGC test jack, J22. S5 shall conform to MS75028

3.7.5.1.9 AUDIO OUTPUT switch, S6 - Switch, S6, shall be a four position rotary switch which shall provide the capability of selecting which one of the four audio outputs is connected to the SELECTED OUTPUT test jack, J17. The Test Set shall load the output lines as follows.

ADF AUDIO OUTPUT	20K ± 2K ohms
X-MODE RCVD AUDIO	500 ± 50 ohms
X-MODE GUARD AUDIO	600 ± 60 ohms
UNATTENUATED AUDIO OUTPUT	No Load

3.7.5.1.10 AUDIO INPUT switch, S7 - Switch, S7, shall be a five position rotary switch which shall provide the capability of selecting which one of five audio inputs is connected to the SELECTED AUDIO INPUT test jack, J21 The five inputs are X-Mode Received Audio Input, X-Mode Transmit Audio, Retransmit Audio, CASS, and Unattenuated Audio Input.

3.7.5.1 11 PRESET CHAN switch, S8 - Switch, S8, shall be a twenty position rotary switch which shall provide the control of the preset channel number portion of the frequency data control word generated by the Test Set internal data generator.

3.7.5.1 12 MODE switch, S9 - Switch, S9, shall be a four position rotary switch which shall provide the capability to control the mode output of the data generator internal to the Test Set The four positions shall be identified GUARD, MANUAL, PRESET, and READ.

3.7.5 1 13 TAKE CMD switch, S11 - Switch, S11, shall be a two position toggle switch which shall apply a ground either to the TAKE CONTROL line of the Panel Radio or Remote R/T or the REMOTE TAKE CONTROL line of the Remote Control unit. In addition, S10 in conjunction with the TESTED UNIT switch, S12, shall determine when the TEST SET provides simulated frequency control of the unit(s) under test. Control by the Test Set shall be removed by placing the GUARD PRECED switch in the ON position S11 shall conform to MS75029.

3.7.5.1 14 TESTED UNIT switch, S12 - Switch, S12, shall be a six position rotary switch which shall be active only in the AN/ARC-159(V) position of the EQPT TYPE switch, S13. S12 shall select the inputs, outputs, and controls of the Test Set which are switched or enabled to the unit(s) under test as described elsewhere herein. The positions of the switch shall be labeled: PNL RAD ONLY, RMT RT ONLY, RMT CONT ONLY, RMT IND ONLY, PNL RAD/RMT CONT and RMT RT/RMT CONT

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3.7.5.1.15 EQPT TYPE switch, S13 - Switch, S13, shall be a two position rotary switch. The switch shall enable or disable Test Set controls, inputs, and outputs as appropriate to test the unit type under test. The Test Set shall contain circuitry to verify that the unit(s) under test connected to the Test Set and the position of S13 are compatible If not, the primary power to the unit(s) under test shall be interrupted.

3.7.5.1.16 X-MODE AUDIO SEL switch, S14 - Switch, S14, shall be a toggle switch which shall be capable of placing a ground on the AUX-MODE AUDIO SELECTOR GROUND line of the unit under test. S14 shall conform to MS75028.

3.7.5.1.17 MAIN SQUELCH switch, S15 - Switch, S15, shall be a toggle switch which shall be capable of placing a ground on the MAIN SQUELCH ON-OFF line of the unit under test. The switch shall be operative only in the RMT RT ONLY position of the TESTED UNIT switch, S12. S15 shall conform to MS75028.

3.7.5.1.18 ADF ENBL switch, S16 - Switch, S16, shall be a toggle switch which shall be capable of placing a ground on the ADF ENABLE line of the unit under test. This switch shall be operative only in the RMT RT ONLY and RMT RT/RMT CONT positions of the TESTED UNIT switch, S12. S16 shall conform to MS75028.

3.7.5.1.19 TONE XMT CONT switch, S17 - Switch, S17, shall be a momentary toggle switch which shall be capable of placing a ground on the TONE TRANSMIT CONTROL line of the unit under test. This switch shall be operative only in the RMT RT ONLY and RMT RT/RMT CONT positions of the TESTED UNIT switch, S12. S17 shall conform to MS75028

3.7.5.1.20 BW CONT switch, S18 - Switch, S18, shall be a toggle switch that shall be capable of placing a ground on the AUX-MODE CONTROL line of the unit under test S18 shall be operative only in the AN/ARC-159 position of the EQPT TYPE switch, S13, or the PNL RAD ONLY, RMT RT/RMT CONT and RMT RT ONLY positions of the TESTED UNIT switch, S12. S18 shall conform to MS75028

3.7.5.1.21 POWER CONT switch, S19 - Switch, S19, shall be a toggle switch which shall be capable of placing a ground on the POWER ON-OFF line of the unit under test. S19 shall conform to MS75028

3.7.5.1.22 FREQUENCY switch, S20 - Switch, S20, shall be a twelve position rotary switch which shall control the units and tens of kilohertz frequency control data generated by the Test Set internal data generator. The switch positions shall select 00, 25, 50 and 75 kilohertz control conditions three times in one revolution of the switch The switch shall be operative only in the MANUAL, PRESET, and READ positions of the MODE switch, S9.

3.7.5.1.23 FREQUENCY switch, S21 - Switch, S21, shall be a ten position rotary switch which shall control the hundreds of kilohertz frequency control data generated by the Test Set internal data generator The switch shall be operative only in the MANUAL, PRESET and READ positions of the MODE switch, S9

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3.7.5.1.24 FREQUENCY switch, S22 - Switch, S22, shall be identical in function to switch, S21, with the exception that S22 shall control the units of megahertz frequency control data generated by the Test Set internal data generator.

3.7.5.1.25 FREQUENCY switch, S23 - Switch, S23, shall be an eighteen position rotary switch which shall control the tens and hundreds of megahertz frequency control data generated by the Test Set internal data generator providing digits from 22 to 39. The switch shall be operative only in the MANUAL, PRESET, and READ positions of the MODE SELECTOR switch, S9.

3.7.5.1.26 XMT CONT (KEY) switch, S10 - Switch, S10 shall be a toggle switch which shall be capable of applying a ground on the TRANSMIT CONTROL line of the unit under test. S10 shall not be operative in the RMT CONT ONLY and RMT IND ONLY positions of the TESTED UNIT switch, S12

3.7.5.1.27 IND TEST switch, S24 - Switch, S24, shall be a momentary push-button switch that shall provide a press to test function such that when depressed all segments of the six FREQUENCY digital displays shall illuminate. S24 shall conform to MIL-S-8805.

3.7.6 Electrical connections - Connections to external circuits shall be provided as follows

Designation	Type	Function
J1	MS3102R-28-S1S	Mates Cable W1
J2	MS3102R-22-14P	Mates Cable W2
J3	MS3102R-24-28S	Mates Cable W3
J4	MS3102R-20-27S	Mates Cable W4
J5	MS39024/10-03	Mates low side of external DC voltmeter to primary Tested Unit power line
J6	MIL-C-39024	Mates high side of external DC voltmeter to primary Tested Unit power line
J7	MIL-C-39024	Mates low side of external DC voltmeter to 27.5V Sw output
J8	MIL-C-39024	Mates high side of external DC voltmeter to 27.5V Sw output
J9	MIL-C-39024	Mates low side of external DC voltmeter to REXMT CONT OUT line

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3.7.6 Electrical connections, continued

Designation	Type	Function
J10	MIL-C-39024	Mates high side of external DC voltmeter to REXMT CONT OUT line
J11	MIL-C-39024	Mates low side of oscilloscope to DATA B
J12	MIL-C-39024	Mates high side of oscilloscope to DATA A
J13	MIL-C-39024	Mates low side of oscilloscope to CLOCK B
J14	MIL-C-39024	Mates high side of oscilloscope to CLOCK A
J15	MIL-C-39024	Mates oscilloscope trigger input to Data Word Sync line
J16	Dual Binding Post	Mates audio voltmeter to Headset output
J17	Dual Binding Post	Mates audio voltmeter to Selected Audio Output
J18	Dual Binding Post	Mates audio voltmeter to output of the MIC Adapter
J19	Dual Binding Post	Mates audio signal generator to input of the MIC Adapter
J20	Dual Binding Post	Mates audio voltmeter to output of test set audio amplifier
J21	Dual Binding Post	Mates audio signal generator to input of test set audio amplifier
J22	Dual Binding Post	Mates Vacuum Tube Voltmeter to selected IF AGC line
J23	Banana Tip Jack, Black	Mates ammeter calibrator to (-) side of M1
J24	Banana Tip Jack, Black	Mates ammeter calibrator to (+) side of M1

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3.7.7 Enclosure - The chassis and components on the rear of the Test Set panel shall be protected by an enclosure. The enclosure shall be designed such that the Test Set can rest on any of its six sides without distortion of the front panel or chassis and without damage to any Test Set component.

3.7.8 Cable assemblies - Cable assemblies shall be provided as follows:

(1) Cable W1 - Cable W1 shall interconnect the Test Set connector, J1, with either the AN/ARC-159 connector, J2, the AN/ARC-159(V) Radio Set connector, J3, or the RT/1150/ARC-159(V) Receiver-Transmitter, Radio Connector, J3. W1 shall be 36 ± 6.0 inches in length and shall contain the following connectors

Reference Designation	Connector Type
P1	MS3106A-26-21P (with MS3057-16A Clamp)
P2	MS27473T18B-32S (with MS27506B18-2 Strain Relief)

(2) Cable W2 - Cable W2 shall interconnect the Test Set connector, J2, with either the AN/ARC-159(V) Radio Set connector, J4 or the RT/1150/ARC-159(V) Receiver-Transmitter, Radio Connector, J4. W2 shall be 36 ± 6.0 inches in length and shall contain the following connectors

Reference Designation	Connector Type
P1	MS3106A-22-14S (with MS3057-12A Clamp)
P2	MS2743T12F35P (with MS27506B12-2 Strain Relief)

(3) Cable W3 - Cable W3 shall interconnect the Test Set connector, J3, with the C-9577/ARC-159(V) Control, Radio Set connector, J1. W3 shall be 36 ± 6.0 inches in length and shall contain the following connectors

Reference Designation	Connector Type
P1	MS3106A-24-28P (with MS3057-16A Clamp)
P2	MS27484T18B32S (with MS27506B18-2 Strain Relief)

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3.7.8 Cable assemblies, continued

(4) Cable W4 - Cable W4 shall interconnect the Test Set connector, J4, with the ID-1972/ARC-159(V) Indicator, Frequency Channel connector, J1. W4 shall be 36 ± 6.0 inches in length and shall contain the following connectors.

Reference Designation	Connector Type
P1	MS3106A-20-27P (with MS3057-12A Clamp)
P2	MS27484T10F35S (with MS27506F10-2 Strain Relief)

3.7.9 Meters and indicators - The Test Set shall have the following meters and indicators.

3.7.9.1 AMMETER, M1 - Ammeter M1 shall monitor the primary power DC current required by the unit(s) under test. M1 shall have a range of 0 - 10 amperes and shall conform to MIL-M-10304.

3.7.9.2 TEST SET 28VDC indicator, DS-1 - Indicator, DS-1, shall illuminate whenever circuit breaker, CB1, is turned ON and power is supplied to the Test Set. DS-1 shall conform to MIL-L-3661 and shall contain a green lens

3.7.9.3 TESTED UNIT 28VDC indicator, DS-2 - Indicator, DS-2, shall illuminate whenever circuit breaker, CB2, is turned ON and power is supplied to the unit(s) under test. DS-2 shall conform to MIL-L-3661 and shall contain a green lens.

3.7.9 4 PNL LAMP 28VDC indicator, DS-3 - Indicator, DS-3, shall illuminate whenever circuit breaker, CB3, is turned ON and power is supplied to the unit(s) under test 28V lamp circuit. DS-3 shall confrom to MIL-L-3661 and shall contain a green lens.

3.7.9.5 PNL LAMP 5VDC indicator, DS-4 - Indicator, DS-4, shall illuminate whenever switch, S1, is turned ON and power is supplied to the unit(s) under test 5V lamp circuit DS-4 shall conform to MIL-L-3661 and shall contain a green lens

3.7.9.6 GUARD RCVR indicator, DS-5 - Indicator, DS-5, shall illuminate whenever the GUARD RECEIVER ON-OFF line is at ground potential whether the ground is supplied by the unit under test or by the Test Set GUARD RCVR switch, S4 DS-5 shall conform to MIL-L-3661 and shall contain a white lens

3.7.9.7 TAKE CMD indicator, DS-7 - Indicator, DS-7, shall illuminate whenever the Test Set has taken command of the frequency and mode control of the unit under test DS-7 shall conform to MIL-L-3661 and shall contain a white lens

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3.7.9.8 X-MODE AUDIO SEL indicator, DS-8 - Indicator, DS-8, shall illuminate whenever the X-MODE AUDIO SELECTOR GROUND line is placed at ground potential by the X-MODE AUDIO SEL switch, S14. DS-8 shall conform to MIL-L-3661 and shall contain a white lens.

3.7.9.9 MAIN SQUELCH DISABLE indicator, DS-9 - Indicator, DS-9, shall illuminate whenever the MAIN SQUELCH ON-OFF line is placed at ground potential whether the ground is supplied by the unit under test or by the MAIN SQUELCH switch, S15. DS-9 shall conform to MIL-L-3661 and shall contain a white lens.

3.7.9.10 ADF ENBL indicator, DS-10 - Indicator, DS-10 shall illuminate whenever the ADF ENABLE line is placed at ground potential whether the ground is supplied by the unit under test or by the ADF ENBL switch, S16. DS-10 shall conform to MIL-L-3661 and shall contain a white lens.

3.7.9.11 TONE XMT CONT indicator, DS-11 - Indicator, DS-11, shall illuminate whenever the TONE TRANSMIT CONTROL line is placed at ground potential whether the ground is supplied by the unit under test or by the TONE XMT CONT switch, S17. DS-11 shall conform to MIL-L-3661 and shall contain a white lens

3.7.9.12 BW CONT indicator, DS-12 - Indicator, DS-12, shall illuminate whenever the AUX MODE CONTROL line is placed at ground potential whether the ground is supplied by the unit under test or by the BW CONT switch, S18. DS-12 shall confrom to MIL-L-3661 and shall contain a white lens

3.7.9.13 POWER CONT indicator, DS-13 - Indicator, DS-13, shall illuminate whenever the POWER ON-OFF line is placed at ground potential whether the ground is supplied by the unit under test or by the POWER CONT switch, S19 DS-13 will conform to MIL-L-3661 and shall contain a white lens.

3.7.9.14 XMT CONT (KEY) indicator, DS-6 - Indicator, DS-6, shall illuminate whenever a ground is placed on the TRANSMIT CONTROL (PTT) line by the XMT CONT (KEY) switch, S10 DS-6 shall conform to MIL-L-3661 and shall contain an amber lens.

3.7.10 Test Set internal functional assemblies - The Test Set shall contain the following internal functional capabilities Each functional capability shall not be required to be a separate assembly.

3.7.10.1 Digital display - The six digit display shall either indicate the selected frequency in megahertz or the center two digits shall indicate the selected preset channel in the PRESET mode. The display shall indicate the information generated by the unit under test or the Test Set depending upon which unit has control of the frequency data control line The digital display indicators shall be of the seven segment plug-in type.

3.7.10.2 Power supply - The Test Set power supply shall supply the DC voltages required to operate all of the functional capabilities of the Test Set The power supply shall provide satisfactory operation from the primary DC power line with the characteristics as specified in MIL-STD-704.

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3.7.10.3 Clock - The Test Set clock shall provide a two line output to the data frequency control circuitry of the AN/ARC-159(V) equipment under test or of the Test Set as required. The clock frequency shall be nominally 800Hz. The two clock lines shall provide signals of opposite phase. The amplitudes of each clock signal shall nominally be -8V and +2V. The clock signal shall be on for 24 bits and off for 8 bits. The rise and fall times of the clock signals shall be 220 ± 50 usec. The output impedance of the clock shall be nominally 300 ohms.

3.7.10.4 Data generator - The Test Set data generator shall supply the frequency data control information to the unit(s) under test whenever the Test Set is in command of the frequency control. The generator shall provide a data word length of 32 bits with 24 bits of information and 8 bits of silence for synchronizing purposes. The bit allocation is as follows:

Bit #	Function	Bit #	Function
1	Mode A	13	8 MHz
2	Channel 1	14	10 MHz
3	Channel 2	15	20 MHz
4	Channel 4	16	40 MHz
5	Channel 8	17	80 MHz
6	Channel 10	18	100 KHz
7	Channel 20	19	200 KHz
8	Mode B	20	400 KHz
9	Mode A	21	800 KHz
10	1 MHz	22	25 KHz
11	2 MHz	23	50 KHz
12	4 MHz	24	300 MHz

Mode A and Mode B determine what the indicator will display according to Table 1

Mode	A	B
MANUAL	1	0
PRESET READ	1	1
PRESET	0	0
GUARD	0	1

Table 1 Indicator Display Determination

The frequency of data generation shall be controlled by the clock. The generator shall provide a two line output with signals of opposite phase. The amplitudes of the signals shall be nominally -8V and +2V. The output impedance shall be nominally 300 ohms.

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3.7.10.4 Data generator, continued

The data generator shall be controlled by the FREQUENCY switches S20 through S23, the MODE switch, S9, and the PRESET CHAN switch, S8. In the GUARD mode the frequency shall be controlled by a fixed diode matrix in the Test Set providing a frequency of 243.000 MHz. The data generator shall be enabled under the following switch conditions: when the TESTED UNIT switch, S12, is in the RMT RT ONLY and RMT IND ONLY positions, when S12 is in the RMT CONT ONLY and RMT RT/RMT CONT positions and the TAKE CMD switch, S11 is in the RADIO position, and when S12 is in the PNL RAD ONLY position and S11 is in the RMT CONT position The data generator shall be disabled when the GUARD PRECED switch, S2, is in the ON position.

3.7.10.5 Data receiver - The data receiver shall process the frequency control data generated either by the Test Set or the unit under test and drive the FREQUENCY display.

3.7.10.6 Module extenders - The Test Set shall contain the following module extenders:

(1) Power Amplifier/Power Supply Module Extender

(2) Control Module Extender

(3) Frequency Synthesizer Module Extender

(4) Translator Module Extender

(5) IF/AF Module Extender

(6) Guard Receiver Module Extender

Each extender shall plug into the AN/ARC-159 Radio Set or the AN/ARC-159(V) Radio Set or the RT/1150/ARC-159(V) Receiver-Transmitter Radio, in place of the module under test and shall accept the module under test. Between the Radio Set adapter and the module adapter the extender shall contain a cable to allow the module to be extended 36 ± 6.0 inches from the radio and to allow the module to be laid on the table surface in any orientation. The extender shall be designed to provide protection of mating connectors and each adapter shall contain a handle to facilitate plugging the extenders into the radio and plugging the module into the extender.

3.7.11 Electrical power source - The test equipment shall operate from a power source as specified in MIL-T-21200. The power required shall be of the following type and shall not exceed the specified amount.

DC Power, 28V, 10 amp.

3.7.11.1 Protection - The equipment shall be capable of withstanding input power variations as imposed by abnormal electric system operations as defined in MIL-STD-704.

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3.7.12 Warm-up time - No warm-up time shall be required prior to operation.

3.8 Dimensions - The maximum overall size shall not exceed 19.0 inches wide by 14.0 inches high by 16.5 inches long not including the pendant power cable, cable assemblies, and module extenders.

3.9 Finish - Finishes shall be in accordance with MIL-T-21200.

3.10 Interchangeability - Interchangeability shall be in accordance with MIL-T-21200.

3.11 Workmanship - Workmanship shall be in accordance with MIL-T-21200.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any inspections deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

First article (preproduction) tests

Initial production tests

Acceptance tests

Life test

4.2 First article (preproduction) tests - First article (preproduction) tests shall be made on an equipment representative of the production equipments to be supplied under the contract The tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4 6 The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 First article (preproduction) test data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, and ambient and surface temperatures The ambient and surface temperature shall be obtained under the extreme high temperature operating condition

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4.2.2 Scope of tests - The tests shall include all tests deemed necessary, by the procuring activity, to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Within the limits specified in MIL-T-21200, first article tests shall include environmental tests in accordance with MIL-T-5422 and interference test in accordance with MIL-STD-462 to the MIL-STD-461 requirements.

4.2.2.1 Maintainability assurance test - A maintainability assurance test shall be conducted in accordance with MIL-STD-471 to demonstrate the requirements of 3.4.2. The test details shall include as a minimum, task selection, test method, demonstration support material, demonstration conditions, demonstration test team, test schedule and test reports, and shall be a part of the test procedure, in accordance with 4.6, to be submitted to the procuring activity for approval.

4.2.2.2 Probability of successful performance test - A probability of successful performance assurance test to demonstrate the requirement of 3.4.3 shall be conducted. The test procedure shall be prepared in accordance with 4.6 and submitted by the contractor to the procuring activity for approval.

4.2.3 First article (preproduction) approval - Approval of the test sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the test sample. Prefabrication of production equipment prior to the approval of the test sample is at the contractor's own risk. The approved test sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The test sample shall not be considered as one of the equipments under the contract.

4.2.4 Production equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the approved test sample Each equipment shall be capable of successfully passing the same tests as imposed on the test sample Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial production tests - One of the first ten production equipment shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The test sample shall not be selected for this test.

4.3.1 Scope of tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved test sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory material - In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such necessary material and data necessary to test the equipment.

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4.3.3 Initial production sample approval - Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of initial production test sample - On completion of the initial production test, the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall re-submit the equipment for acceptance.

4.4 Acceptance tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product Acceptance tests shall consist of the following

(1)	Individual tests	(4.4.1)
(2)	Sampling tests	(4.4.2)
(3)	Reliability assurance tests	(4 4.3)
(4)	Special tests	(4.4.4)

4.4.1 Individual tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1)	Examination of product	(4.4.1.1)
(2)	Operational test	(4.4.1.2)
(3)	Manufacturing run in test	(4 4.1.3)

4.4 1.1 Examination of product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met

4.4.1.2 Operational test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

NOTE: Approval of the operational test procedure must be obtained from the Naval Air Systems Command prior to testing.

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4.4.1.3 Manufacturing run in test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30Hz (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship Operating within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hours period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling tests - Equipment selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action).

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Tests
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

*NOTE: If by contract action the Initial production test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

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4.4.2.1. Scope of tests - As a minimum each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4 4.3 Reliability assurance tests - Reliability assurance tests shall be conducted using MIL-STD-781. Tests shall be conducted as required by both the Qualification Phase and Sampling Phase.

4.4.3.1 Qualification phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipment to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan 6 shall be used

4.4.3.2 Reliability sampling phase tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the Section entitled "Sampling Phase of Production Reliability Tests" For the Sampling Phase Test Level E shall be used The Accept-Reject criteria for Test Plan 6 shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.4.3.2.1 Lot size for sampling phase - The equipments constructed during one month shall be one lot.

4.4.3.3 Test details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests

4.4.4 Special tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

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4.4.4.1 Special test schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken.

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in failure tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual tests. The life test sample shall be performed under the conditions specified in 4.5.1. The life test shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

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4.5.1 Test conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Room Ambient
Altitude	Normal ground (0-5000 ft.)
Humidity	Room Ambient
AC Voltage	115 ± 5 volts (at lowest applicable frequency)
DC Voltage	27.0 ± 2.0 volts
Vibration	None

4.5.2 Test periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following.

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

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4.5.5 Reconditioning of life test samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of tested equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements

4.9 Rejection and retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the government inspector

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794 In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

5.2 Marking - In addition to any special marking by the contract or order, each unit package and exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use - The equipment shall be for shop use and shall provide the capability to perform functional tests and fault isolation procedures on AN/ARC-159 and AN/ARC-159(V) UHF Radio Set equipments. The Test Set shall provide the capability to perform procedures to fault isolate the major module subassemblies of the Radio Set under test.

6.2 Performance objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-T-21200, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3 Nonrepairable subassemblies - As a general rule, nonrepairable subassemblies should be encapsulated or hermetically-sealed The number of connections internal to the subassembly should be held to a minimum Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.4 Precedence of documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification

(2) This specification - This specification shall have precedence over all applicable subsidiary specifications Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity

(3) Referenced specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified

6.5 Ordering data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following

(1) Title, number and date of this specification

(2) Selection of applicable levels of packaging and packing (See 5.1).

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6.6 Type designation - The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.4.6. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Revisions - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 Associated equipment - The equipment shall operate with the following associated equipment.

Item Name	Federal Mfg. Code	Manufacturer's Part No
Signal Generator	88869	FM2748
Power Meter	24655	ME-2A
Multimeter VTVM	80058	AN/USM-116C
RF Power Meter	80058	AN/URM-43E
GP Oscilloscope	80058	AN/USM-281A
Distortion Analyzer	28480	334A
Attenuator	80058	UG-528/U
Double Balanced Mixer	28480	10514A
Variable Atten.	24655	874GAL
Crystal Detector	28480	420B
Multimeter, VOM	80058	AN/USM-311
Digital Voltmeter	89536	8300A01-02
Decade Resistor	24655	1433-9714 Type 1433M
RF Voltmeter	04901	91H-S7
Sweep Sig Gen	80058	SG-1020/UR
Spectrum Analyzer	80009	491
Pulse Generator	06811	110B
Frequency Counter	28480	H61-5245L
Frequency Counter Converter	28480	5253B
Audio Oscillator	28480	200CD

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6.9 Definitions -

Custodians	Preparing Activity
Navy - AS	Navy - AS
	Project No.

Project No. 6625-N670

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DOCUMENT IDENTIFIER AND TITLE
MIL-T-81998(AS) TEST SET, RADIO SET AN/ARM-165

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

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